

LOWELL, ID, USA

WMO: 725784

Lat: 46.144N

Lon: 115.596W

Elev: 451

StdP: 96.02

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 04109

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-10.4	-7.5	-16.0	1.0	-7.9	-12.4	1.4	-4.1	4.0	-4.1	3.1	-2.6	1.2	0	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	35.0	19.8	32.9	19.0	31.0	18.4	21.1	31.2	20.1	30.2	19.2	28.8	0.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	13.5	24.0	17.0	12.8	23.0	15.9	11.9	22.0	63.1	31.2	59.4	30.4	56.2	28.8	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.7	2.9	2.5	-13.5	37.9	3.3	1.6	-15.8	39.1	-17.7	40.0	-19.5	40.9	-21.9	42.1		
(5)			-13.5	23.3	3.2	0.8	-15.8	23.9	-17.7	24.3	-19.5	24.8	-21.8	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	0.7	2.4	6.2	9.7	14.0	17.8	22.6	21.8	16.8	9.8	4.1	0.3	(6)
(7)		DBStd	8.32	3.28	3.44	2.88	3.15	3.49	3.71	2.80	2.70	3.32	3.26	3.26	3.13	(7)
(8)		HDD10.0	1192	287	212	121	43	6	1	0	0	1	42	179	300	(8)
(9)		HDD18.3	3143	546	445	375	259	139	53	4	6	67	263	428	558	(9)
(10)		CDD10.0	1399	0	0	4	34	131	234	391	366	204	36	1	0	(10)
(11)		CDD18.3	311	0	0	0	0	5	36	136	114	20	0	0	0	(11)
(12)		CDH23.3	3638	0	0	0	18	154	491	1469	1218	284	4	0	0	(12)
(13)		CDH26.7	1693	0	0	0	2	40	188	768	611	84	0	0	0	(13)
(14)	Wind	WSAvg	0.7	0.7	0.7	0.7	0.7	0.8	0.9	0.8	0.7	0.5	0.7	0.7	(14)	
(15)	Precipitation	PrecAvg	827	88	60	77	81	83	80	31	36	51	71	99	79	(15)
(16)		PrecMax	1256	211	148	206	151	172	192	98	112	129	195	287	190	(16)
(17)		PrecMin	483	12	2	13	13	25	18	0	0	2	0	19	5	(17)
(18)		PrecStd	173	48	33	40	34	29	41	27	34	36	47	54	42	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	13.3	19.5	25.9	30.4	34.6	37.8	36.9	32.1	23.4	13.1	7.4	(19)
(20)			MCWB	5.5	8.0	10.9	13.9	16.3	20.2	20.5	19.8	18.9	15.2	9.7	5.9	(20)
(21)		2%	DB	6.2	10.1	16.2	22.3	27.4	31.2	35.8	34.4	29.0	19.1	10.9	6.0	(21)
(22)			MCWB	4.6	6.5	9.4	12.9	15.5	18.6	20.0	19.5	17.3	13.3	8.6	4.7	(22)
(23)		5%	DB	4.6	7.8	13.2	18.9	24.7	28.7	33.6	32.7	26.9	16.6	9.2	4.5	(23)
(24)			MCWB	3.3	5.1	8.1	11.2	14.6	17.3	19.4	18.8	16.6	11.7	7.4	3.6	(24)
(25)		10%	DB	3.5	6.1	10.9	16.0	21.8	26.1	31.5	30.6	24.0	14.0	8.0	3.4	(25)
(26)			MCWB	2.6	4.5	7.2	9.7	13.6	16.2	18.6	18.2	15.5	10.6	6.1	2.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	8.8	12.0	15.2	18.0	21.3	23.0	21.9	20.2	16.2	10.8	6.5	(27)
(28)			MCDWB	7.5	12.1	17.7	23.5	26.7	30.9	32.5	32.7	28.5	21.0	12.1	7.2	(28)
(29)		2%	WB	4.7	6.9	10.0	13.3	16.5	19.7	21.5	20.7	18.7	14.1	9.1	5.0	(29)
(30)			MCDWB	6.0	9.2	14.7	20.6	25.3	29.0	31.4	31.3	27.2	18.0	10.3	5.8	(30)
(31)		5%	WB	3.6	5.7	8.7	11.9	15.4	18.3	20.4	19.8	17.4	12.5	8.0	3.8	(31)
(32)			MCDWB	4.5	7.4	12.6	17.6	22.6	26.5	30.4	30.2	24.9	15.4	9.0	4.4	(32)
(33)		10%	WB	2.8	4.6	7.6	10.5	14.2	17.1	19.5	18.8	16.1	11.4	6.8	2.8	(33)
(34)			MCDWB	3.4	6.0	10.4	14.6	20.3	24.1	29.5	28.7	22.7	13.5	7.8	3.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.0	7.2	10.1	12.1	14.0	15.1	18.9	18.9	15.6	9.9	5.8	4.5	(35)
(36)			MCDBR	6.1	9.9	14.7	18.6	19.8	19.8	21.8	21.5	19.1	13.8	7.6	5.3	(36)
(37)			MCWBR	4.5	6.5	8.5	9.4	8.7	8.2	8.2	8.4	8.9	8.0	5.5	4.0	(37)
(38)		5% WB	MCDBR	5.6	9.2	13.1	16.8	17.4	18.7	19.6	20.0	18.1	12.1	6.9	4.8	(38)
(39)			MCWBR	4.2	6.2	8.1	9.0	8.1	8.2	8.0	8.5	8.8	7.3	5.1	3.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.282	0.287	0.328	0.382	0.407	0.401	0.411	0.455	0.408	0.339	0.339	0.279	(40)	
(41)		taud	2.467	2.463	2.421	2.288	2.271	2.331	2.316	2.155	2.261	2.485	2.343	2.505	(41)	
(42)		Ebn at Noon	832	900	902	874	861	866	851	793	800	823	733	798	(42)	
(43)		Edh at Noon	67	81	99	123	130	123	124	140	115	80	76	58	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.32	2.12	3.25	4.65	5.61	6.43	7.39	6.22	4.51	2.69	1.45	1.01	(44)	
(45)		RadStd	0.16	0.25	0.33	0.39	0.45	0.56	0.41	0.34	0.43	0.27	0.14	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83	+38

Nomenclature: See separate page